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High-Efficiency Forward Modeling of Gravitational Fields in Spherical Harmonic Domain with Application to Lunar Topography Correction

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Abstract: Gravity forward modeling as a basic tool has been widely used for topography correction and 3D density inversion. The source region is usually discretized into tesseroids (i.e., spherical prisms) to consider the influence of the curvature of planets in global or large-scale problems. Traditional gravity forward modeling methods in spherical coordinates, including the Taylor expansion and Gaussian–Legendre quadrature, are all based on spatial domains, which mostly have low computational efficiency. This study proposes a high-efficiency forward modeling method of gravitational fields in the spherical harmonic domain, in which the gravity anomalies and gradient tensors can be expressed as spherical harmonic synthesis forms of spherical harmonic coefficients of 3D density distribution. A homogeneous spherical shell model is used to test its effectiveness compared with traditional spatial domain methods. It demonstrates that the computational efficiency of the proposed spherical harmonic domain method is improved by four orders of magnitude with a similar level of computational accuracy compared with the optimized 3D GLQ method. The test also shows that the computational time of the proposed method is not affected by the observation height. Finally, the proposed forward method is applied to the topography correction of the Moon. The results show that the gravity response of the topography obtained with our method is close to that of the optimized 3D GLQ method and is also consistent with previous results.



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Keywords: gravity forward modeling; spherical harmonic domain; tesseroid; spherical shell; 3D-GLQ; lunar topography

1. Introduction

The forward modeling of gravitational fields is an important step in data processing and geological interpretation in gravity exploration and has been widely used to calculate the gravity responses of topography, sedimentary layers, crystalline layers, and the Moho interface [1–5]. These effects can then be removed from the free-air gravity anomaly to obtain the residual gravity anomaly only produced by the inhomogeneity of the upper mantle of the lithosphere [6–9]. In addition, gravity forward modeling, as the core of iteration of 3D density inversion, is typically employed in mineral resource exploration [10–12] and modeling the lithospheric density structures of planetary interiors [13–15]. However, traditional gravity forward methods are mostly based on the Cartesian coordinate system [16,17]. In recent years, with the development of satellite gravity, an increasing number of high-precision and high-resolution gravity field models have been applied to the study of 3D density structures and the evolutionary history of the Earth [18,19], the Moon [20–22], and Mars [23,24]. Therefore, gravity forward methods and relative methods of data processing and inversion should be changed from the Cartesian coordinate system to spherical coordinates.

Gravity forward modeling in spherical coordinates can be carried out in the spatial and spherical harmonic domains (i.e., frequency domain). In the spatial domain, the source region is typically discretized into spherical tesserooids [25]. The gravity responses of these tesserooids can be obtained through the accumulation of gravity signals generated by each tesserooid. However, the 3D Newton's integral of the tesserooid is usually hard to solve analytically, except in cases where the computation point is situated on the polar axis [26]. Therefore, numerical integration methods, such as the Taylor series expansion [27,28] and the 2D/3D Gauss–Legendre quadrature (GLQ) method, are more common alternatives used to solve this problem [29–31]. However, these numerical methods always lead to incorrect results when the observation points are close to tesserooids. Uieda et al. [32] proposed an adaptive discretization method that greatly increased the computational accuracy with a maximum relative error of less than 0.1%, and the only disadvantage of this method is that the computational efficiency decreases rapidly with the decrease in observation height. In addition, computational efficiency poses another issue for large-scale or global gravity forward modeling and inversion, especially when the discretization is fine. Recently, some strategies have been proposed, e.g., kernel matrix equivalent storage [4,33] and fast kernel–vector multiplication [5,34], increasing the computational efficiency by three orders of magnitude. Though the issues of the computational efficiency and accuracy of these methods in the spatial domain have been resolved, global and regional gravitational fields are now more commonly released in the form of spherical harmonic coefficients, which makes it difficult for these spatial domain methods to simulate gravity responses with specified degrees and orders. However, this will be simple for spherical harmonic domain methods.

Spherical harmonics are special functions defined on the surface of a sphere that form an orthogonal basis set for square-integrable functions, by which non-singular expressions for gravity anomalies and gradients are derived based on the sums of spherical harmonic coefficients of gravitational field models [35,36]. Similar to Parker's formula [37] and the interface iterative inversion method [38] in Cartesian coordinates, Wieczorek and Phillips [39] derived the formula for calculating the gravity anomaly of a single undulating interface with a constant density and the iterative inversion method of density interface in the spherical harmonic domain, which was used for lunar topography correction and the estimation of lunar crustal thickness [40,41]. This method is highly efficient and suitable for gravity correction using high-order and high-resolution terrain models, as well as for the inversion of a single-density interface. However, the binomial series expansion in this method has a great effect on computational accuracy and convergence in inversion. To mitigate this erroneous behavior, Root et al. [42] proposed a correction that can be used for global gravity modeling of the Earth's complete density structure. Instead of implementing the binomial expansion of the product of topographic relief and 2D density distribution as demonstrated by Wieczorek and Phillips [39], Šprlák et al. [43,44] presented an innovative approach to the rigorous modeling of the spherical gravitational potential spectra from a body's volumetric density and geometry by discretizing the integral and using the discrete Fourier transform. In addition to topography corrections with a constant density, these spherical harmonic domain methods have also been widely used in the calculation of gravity disturbances of the lithosphere from crustal models with variable density distributions [2,8,45]. Although spherical harmonic expressions of gravity anomalies have been extensively studied and applied, studies on spherical harmonic expressions of gravity gradients remain comparatively rare. In addition, there are few studies on the gravity forward and inversion modeling of 3D density anomalies by using spherical harmonic methods [46].

To adapt to the requirement of gravity forward modeling and inversion of 3D density anomalies with arbitrary degrees and orders, this study proposes an efficient forward modeling method of the gravitational field in the spherical harmonic domain, in which the gravity anomalies and gradient components are expressed as a form of spherical harmonic synthesis of spherical harmonic coefficients of arbitrary 3D density distributions. Two synthetic models show that the proposed method has improved the computational

efficiency by four orders of magnitude and with a similar level of computational accuracy compared with the optimized 3D GLQ method [4,5]. Finally, we apply this method to calculate the topography correction and Bouguer gravity anomaly of the Moon.

2. Theory and Methods

2.1. Spatial Domain Forward Modeling Method

In spherical coordinates, the global spherical shell is typically discretized into tessaroids [25], and observation points are assumed to be on a spherical surface aligning with the tessaroid mesh, as shown in Figure 1.

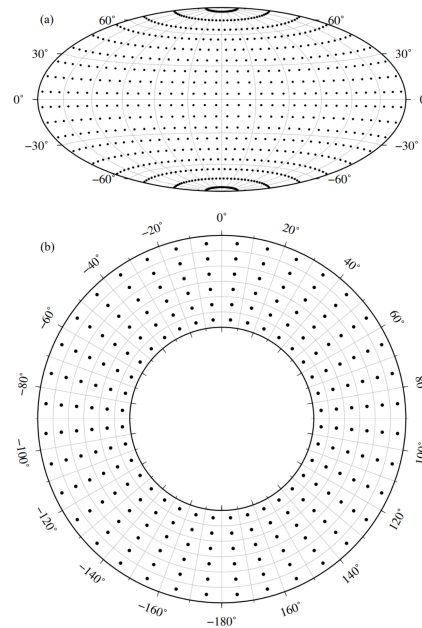


Figure 1. Sketch map of discretization of a global spherical shell (a) and its observation points (b).

Here, we define a local north-oriented frame (LNOF), where the z -axis points upward in the geocentric radial direction, the x -axis points towards the north, and the y -axis points to the east. We suppose that $P(r, \varphi, \lambda)$ is an arbitrary observation point and $Q(r', \varphi', \lambda')$ is the center position of an arbitrary tessaroid bounded by (r_1, r_2) , (φ_1, φ_2) , and (λ_1, λ_2) in three directions, where r , φ , and λ (also for r' , φ' , and λ') are the geocentric radius, latitude, and longitude, respectively. Therefore, the gravitational potential on the observation point P produced by the tessaroid Q with homogeneous density ρ can be expressed as Newton's integral [28,31,32], as follows:

$$V(r, \varphi, \lambda) = G\rho \int_{r_1}^{r_2} \int_{\varphi_1}^{\varphi_2} \int_{\lambda_1}^{\lambda_2} \frac{r'^2 \cos \varphi'}{\ell} dr' d\varphi' d\lambda' \quad (1)$$

where G is the gravitational constant and ℓ is the Euclidean distance between P and Q :

$$\ell = \sqrt{r'^2 + r^2 - 2r'r \cos \psi} \quad (2)$$

$$\cos \psi = \sin \varphi \sin \varphi' + \cos \varphi \cos \varphi' \cos(\lambda' - \lambda)$$

According to the coordinate transformation relationship between the geocentric spherical coordinate system and the local rectangular coordinate system [27,31], it is easy to obtain their partial derivatives of gravitational potential V to r , φ , and λ . Then, the gravitational

accelerations g_α and gradient tensors $g_{\alpha\beta}$ can be obtained via their combinations [47,48], where $\alpha, \beta \in (x, y, z)$:

$$[g_\alpha] = [g_x \quad g_y \quad g_z]^T = \left[\frac{1}{r} \frac{\partial V}{\partial \varphi} \quad \frac{1}{r \cos \varphi} \frac{\partial V}{\partial \lambda} \quad \frac{\partial V}{\partial r} \right]^T \quad (3)$$

$$[g_{\alpha\beta}] = \begin{bmatrix} \frac{1}{r^2} \left(\frac{\partial^2 V}{\partial \varphi^2} + r \frac{\partial V}{\partial r} \right) & \frac{1}{r^2 \cos \varphi} \left(\frac{\partial^2 V}{\partial \varphi \partial \lambda} + \tan \varphi \frac{\partial V}{\partial \lambda} \right) & \frac{1}{r} \left(\frac{\partial^2 V}{\partial \varphi \partial r} - \frac{1}{r} \frac{\partial V}{\partial \varphi} \right) \\ \frac{1}{r^2 \cos \varphi} \left(\frac{\partial^2 V}{\partial \varphi \partial \lambda} + \tan \varphi \frac{\partial V}{\partial \lambda} \right) & \frac{1}{r^2 \cos^2 \varphi} \left(\frac{\partial^2 V}{\partial \lambda^2} + r \cos^2 \varphi \frac{\partial V}{\partial r} - \cos \varphi \sin \varphi \frac{\partial V}{\partial \varphi} \right) & \frac{1}{r \cos \varphi} \left(\frac{\partial^2 V}{\partial r \partial \lambda} - \frac{1}{r} \frac{\partial V}{\partial \lambda} \right) \\ \frac{1}{r} \left(\frac{\partial^2 V}{\partial \varphi \partial r} - \frac{1}{r} \frac{\partial V}{\partial \varphi} \right) & \frac{1}{r \cos \varphi} \left(\frac{\partial^2 V}{\partial r \partial \lambda} - \frac{1}{r} \frac{\partial V}{\partial \lambda} \right) & \frac{\partial^2 V}{\partial r^2} \end{bmatrix} \quad (4)$$

The gravitational potential, gravitational accelerations, and gradient tensors can be further rearranged as follows [32]:

$$V(r, \varphi, \lambda) = G\rho \int_{r_1}^{r_2} \int_{\varphi_1}^{\varphi_2} \int_{\lambda_1}^{\lambda_2} \frac{1}{\ell} \kappa \cdot dr' d\varphi' d\lambda', \quad (5)$$

$$g_\alpha(r, \varphi, \lambda) = G\rho \int_{r_1}^{r_2} \int_{\varphi_1}^{\varphi_2} \int_{\lambda_1}^{\lambda_2} \frac{\Delta_\alpha}{\ell^3} \kappa \cdot dr' d\varphi' d\lambda', \quad (6)$$

$$g_{\alpha\beta}(r, \varphi, \lambda) = G\rho \int_{r_1}^{r_2} \int_{\varphi_1}^{\varphi_2} \int_{\lambda_1}^{\lambda_2} \left(\frac{3\Delta_\alpha \Delta_\beta}{\ell^5} - \frac{\delta_{\alpha\beta}}{\ell^3} \right) \kappa \cdot dr' d\varphi' d\lambda', \quad (7)$$

where $\delta_{\alpha\beta}$ is Kronecker's delta ($\delta_{\alpha\beta} = 1$ if $\alpha = \beta$, and $\delta_{\alpha\beta} = 0$ if $\alpha \neq \beta$), and

$$\begin{aligned} \Delta_x &= r' [\cos \varphi \sin \varphi' - \sin \varphi \cos \varphi' \cos(\lambda' - \lambda)] \\ \Delta_y &= r' \cos \varphi \sin(\lambda' - \lambda) \\ \Delta_z &= r' \cos \varphi - r \\ \kappa &= r'^2 \cos \varphi' \end{aligned} \quad (8)$$

There is no analytical solution for Equations (5)–(7) since they contain elliptical integrals about φ' and λ' . Therefore, numerical integration methods should be adopted to solve them, e.g., the 3D GLQ and Taylor series expansion methods. Since the 3D GLQ method is numerically better than the Taylor series expansion method [31], we choose the 3D GLQ to represent the spatial domain method. For simplicity, taking one-dimensional GLQ as an example, a line integral of continuous function $f(x)$ in the $[a, b]$ interval can be expressed as follows:

$$\int_a^b f(x) dx \approx \frac{b-a}{2} \sum_{k=0}^N W_k f\left(\frac{b-a}{2} x_k + \frac{a+b}{2}\right) \quad (9)$$

where N is the number of Gaussian nodes; x_k and W_k are the k -th Gaussian node and Gaussian coefficient in the $[1, 1]$ interval, respectively, where the commonly used Gaussian nodes and coefficients are given by Wild-Pfeiffer [31]. In the following comparison, we use two Gaussian nodes in each direction to fit the trade-off between computational accuracy and efficiency.

We note that if we simply use this 3D GLQ method for gravity forward modeling of tesseroids, its computational efficiency and accuracy will be very low, especially when the observation point is close to the surface of the tesseroids and when the source region is discretized finely. Therefore, we will employ the optimal methods described by Uieda et al. [32] and Zhao et al. [5] for comparison with the newly proposed method in the next section.

2.2. Forward Method in Spherical Harmonic Domain

The gravitational field outside the Earth satisfies the Laplace equation, so it can also be expressed as spherical harmonic expansion. Based on this theory, in this section we first derive the spherical harmonic expressions of the gravitational field of a single-layer spherical shell model, and then the expressions of the 3D source region are obtained through the accumulation of each layer. The gravitational potential in Equation (1) is rewritten as follows:

$$V(r, \varphi, \lambda) = G \iint_S \int_{r'} \frac{\rho(r', \varphi', \lambda')}{\ell(r, \varphi, \lambda; r', \varphi', \lambda')} dS dr' \quad (10)$$

where S denotes the range of the spherical integral.

The gravitational potential $V(r, \varphi, \lambda)$ and 3D density distribution $\rho(r', \varphi', \lambda')$ in Equation (10) can be expressed in the form of normalized spherical harmonic functions according to the spherical harmonic expansion theory [49]:

$$V(r, \varphi, \lambda) = \sum_{l=0}^L \sum_{m=0}^l \left[{}^1\bar{V}_l^m(r) \bar{R}_l^m(\varphi, \lambda) + {}^2\bar{V}_l^m(r) \bar{T}_l^m(\varphi, \lambda) \right] \quad (11)$$

$$\rho(r', \varphi', \lambda') = \sum_{l=0}^L \sum_{m=0}^l \left[{}^1\bar{\rho}_l^m(r') \bar{R}_l^m(\varphi', \lambda') + {}^2\bar{\rho}_l^m(r') \bar{T}_l^m(\varphi', \lambda') \right] \quad (12)$$

where ${}^1\bar{V}_l^m(r)$ and ${}^2\bar{V}_l^m(r)$ are fully normalized spherical harmonic coefficients of the gravitational potential with degree l and order m on a sphere surface with radius r ; ${}^1\bar{\rho}_l^m(r')$ and ${}^2\bar{\rho}_l^m(r')$ are fully normalized spherical harmonic coefficients of the 2D density distribution on a layer with radius r' ; L is the maximum spherical harmonic degree; and $\bar{R}_l^m(\varphi, \lambda)$ and $\bar{T}_l^m(\varphi, \lambda)$ are fully normalized spherical harmonic functions, which can be written as [49]:

$$\begin{aligned} \bar{R}_l^m(\varphi, \lambda) &= \bar{P}_l^m(\sin \varphi) \cos m\lambda \\ \bar{T}_l^m(\varphi, \lambda) &= \bar{P}_l^m(\sin \varphi) \sin m\lambda \end{aligned} \quad (13)$$

where $\bar{P}_l^m(\sin \varphi)$ is the 4π fully-normalized associated Legendre function.

The Green function $(1/\ell)$ in Equation (10) is the key to spherical harmonic domain methods which can be expressed as follows [49]:

$$\frac{1}{\ell} = \frac{1}{r} \sum_{l=0}^{\infty} \left(\frac{r'}{r} \right)^l \frac{1}{2l+1} \sum_{m=0}^l \left[\bar{R}_l^m(\varphi, \lambda) \bar{R}_l^m(\varphi', \lambda') + \bar{T}_l^m(\varphi, \lambda) \bar{T}_l^m(\varphi', \lambda') \right], \quad r > r' \quad (14)$$

We should note that the infinite series in Equation (14) only converge uniformly for any observation points that are $r > r'$, i.e., the observation points need to be outside of the source region. Substituting Equations (11), (12), and (14) into (10), and using the orthogonal characteristic of spherical harmonic function, we obtain the following:

$$\begin{aligned} {}^1\bar{V}_l^m(r) &= G \int_{r'} \frac{4\pi r'}{2l+1} \left(\frac{r'}{r} \right)^{l+1} {}^1\bar{\rho}_l^m(r') dr' \\ {}^2\bar{V}_l^m(r) &= G \int_{r'} \frac{4\pi r'}{2l+1} \left(\frac{r'}{r} \right)^{l+1} {}^2\bar{\rho}_l^m(r') dr' \end{aligned} \quad (15)$$

Equation (15) shows that the spherical harmonic coefficients of gravitational potential can be obtained via the radial integration of the spherical harmonic coefficients of 3D density distribution, which is the core of gravity forward modeling in the spherical harmonic domain. By substituting Equation (15) into (10), we can obtain the spherical

harmonic expression of gravitational potential on a sphere with a given radius r outside the source region:

$$V(r, \varphi, \lambda) = 4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{r'}{2l+1} \left(\frac{r'}{r}\right)^{l+1} \left[{}^1\bar{\rho}_l^m(r') \bar{R}_l^m(\varphi, \lambda) + {}^2\bar{\rho}_l^m(r') \bar{T}_l^m(\varphi, \lambda) \right] dr' \quad (16)$$

According to Equations (3) and (4), we can obtain the spherical harmonic expressions of gravitational acceleration and gradient tensors:

$$g_x(r, \varphi, \lambda) = 4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{1}{2l+1} \left(\frac{r'}{r}\right)^{l+2} \left[{}^1\bar{\rho}_l^m(r') \cos m\lambda + {}^2\bar{\rho}_l^m(r') \sin m\lambda \right] \frac{\partial \bar{P}_l^m(\sin \varphi)}{\partial \varphi} dr' \quad (17)$$

$$g_y(r, \varphi, \lambda) = 4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{1}{(2l+1) \cos \varphi} \left(\frac{r'}{r}\right)^{l+2} m \left[-{}^1\bar{\rho}_l^m(r') \sin m\lambda + {}^2\bar{\rho}_l^m(r') \cos m\lambda \right] \bar{P}_l^m(\sin \varphi) dr' \quad (18)$$

$$g_z(r, \varphi, \lambda) = -4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{l+1}{2l+1} \left(\frac{r'}{r}\right)^{l+2} \left[{}^1\bar{\rho}_l^m(r') \bar{R}_l^m(\varphi, \lambda) + {}^2\bar{\rho}_l^m(r') \bar{T}_l^m(\varphi, \lambda) \right] dr' \quad (19)$$

$$g_{xx}(r, \varphi, \lambda) = 4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{1}{r(2l+1)} \left(\frac{r'}{r}\right)^{l+2} \left[{}^1\bar{\rho}_l^m(r') \cos m\lambda + {}^2\bar{\rho}_l^m(r') \sin m\lambda \right] \cdot \left[\frac{\partial^2 \bar{P}_l^m(\sin \varphi)}{\partial \varphi^2} - (l+1) \bar{P}_l^m(\sin \varphi) \right] dr' \quad (20)$$

$$g_{xy}(r, \varphi, \lambda) = 4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{1}{r(2l+1)} \left(\frac{r'}{r}\right)^{l+2} m \left[-{}^1\bar{\rho}_l^m(r') \sin m\lambda + {}^2\bar{\rho}_l^m(r') \cos m\lambda \right] \cdot \left[\frac{\partial \bar{P}_l^m(\sin \varphi)}{\cos \varphi \partial \varphi} + \frac{\tan \varphi}{\cos \varphi} \bar{P}_l^m(\sin \varphi) \right] dr' \quad (21)$$

$$g_{xz}(r, \varphi, \lambda) = -4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{l+2}{r(2l+1)} \left(\frac{r'}{r}\right)^{l+2} \left[{}^1\bar{\rho}_l^m(r') \cos m\lambda + {}^2\bar{\rho}_l^m(r') \sin m\lambda \right] \frac{\partial \bar{P}_l^m(\sin \varphi)}{\partial \varphi} dr' \quad (22)$$

$$g_{yy}(r, \varphi, \lambda) = -4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{1}{r(2l+1)} \left(\frac{r'}{r}\right)^{l+2} \left[{}^1\bar{\rho}_l^m(r') \cos m\lambda + {}^2\bar{\rho}_l^m(r') \sin m\lambda \right] \cdot \left[m^2 \frac{\bar{P}_l^m(\sin \varphi)}{\cos^2 \varphi} + (l+1) \bar{P}_l^m(\sin \varphi) + \frac{\tan \varphi \partial \bar{P}_l^m(\sin \varphi)}{\partial \varphi} \right] \frac{\partial \bar{P}_l^m(\sin \varphi)}{\partial \varphi} dr' \quad (23)$$

$$g_{yz}(r, \varphi, \lambda) = -4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{l+2}{r(2l+1)} \left(\frac{r'}{r}\right)^{l+2} m \left[-{}^1\bar{\rho}_l^m(r') \sin m\lambda + {}^2\bar{\rho}_l^m(r') \cos m\lambda \right] \frac{\bar{P}_l^m(\sin \varphi)}{\cos \varphi} dr' \quad (24)$$

$$g_{zz}(r, \varphi, \lambda) = 4\pi G \sum_{l=0}^L \sum_{m=0}^l \int_{r'} \frac{(l+1)(l+2)}{r'(2l+1)} \left(\frac{r'}{r}\right)^{l+3} \left[{}^1\bar{\rho}_l^m(r') \cos m\lambda + {}^2\bar{\rho}_l^m(r') \sin m\lambda \right] \bar{P}_l^m(\sin \varphi) dr' \quad (25)$$

Equations (17)–(25) contain the singular terms of $1/\cos \varphi$ and $1/\cos^2 \varphi$ when φ is closing to the pole. In addition, some equations contain the terms of first- and second-order derivatives of $\bar{P}_l^m(\sin \varphi)$. These mathematical problems have been solved in previous studies by using non-singular derivatives [35,36,50], so we will not focus on them here.

Equations (17)–(25) above are only theoretical expressions in the form of continuous radial integrals. It is difficult to calculate the spherical harmonic coefficients of a continuous density distribution in actual forward modeling. Therefore, the 3D spherical shell model will be divided into several density layers in the depth direction, and each density layer will be approximated using numerical integration. Finally, the gravitational fields generated by all layers will be summed. Supposing that a spherical shell model with a radius interval of

$[r_a, r_b]$ is evenly divided into N_z layers with the thickness of each layer $\Delta r = (r_b - r_a)/N_z$, the center radius of each layer of the spherical shell is

$$r'_k = r_a + (k - 0.5) \times \Delta r, \quad k = 1, 2, \dots, N_z. \quad (26)$$

Though the 3D source region has been divided into multi-layer spherical shells, ${}^1\bar{\rho}_l^m(r')$ and ${}^2\bar{\rho}_l^m(r')$ are still continuous functions of r' . Assuming that the radial subdivision interval Δr is small enough, then the density distribution on the k -th layer can be seen as a constant equal to its center density. Therefore, the above equations can be numerically solved, taking the commonly used V , g_z , and g_{zz} components as examples:

$$V(r, \varphi, \lambda) = 4\pi G \sum_{k=1}^{N_z} \sum_{l=0}^L \sum_{m=0}^l \frac{[{}^1\bar{\rho}_l^m(r'_k)\bar{R}_l^m(\varphi, \lambda) + {}^2\bar{\rho}_l^m(r'_k)\bar{T}_l^m(\varphi, \lambda)]}{(2l+1)(l+3)r^{l+1}} \left[\left(r'_k + \frac{1}{2}\Delta r\right)^{l+3} - \left(r'_k - \frac{1}{2}\Delta r\right)^{l+3} \right] \quad (27)$$

$$g_z(r, \varphi, \lambda) = -4\pi G \sum_{k=1}^{N_z} \sum_{l=0}^L \sum_{m=0}^l \frac{(l+1)[{}^1\bar{\rho}_l^m(r'_k)\bar{R}_l^m(\varphi, \lambda) + {}^2\bar{\rho}_l^m(r'_k)\bar{T}_l^m(\varphi, \lambda)]}{\left[\left(r'_k + \frac{1}{2}\Delta r\right)^{l+3} - \left(r'_k - \frac{1}{2}\Delta r\right)^{l+3} \right]} \times \quad (28)$$

$$g_{zz}(r, \varphi, \lambda) = 4\pi G \sum_{k=1}^{N_z} \sum_{l=0}^L \sum_{m=0}^l \frac{(l+1)(l+2)[{}^1\bar{\rho}_l^m(r'_k)\bar{R}_l^m(\varphi, \lambda) + {}^2\bar{\rho}_l^m(r'_k)\bar{T}_l^m(\varphi, \lambda)]}{\left[\left(r'_k + \frac{1}{2}\Delta r\right)^{l+3} - \left(r'_k - \frac{1}{2}\Delta r\right)^{l+3} \right]} \times \quad (29)$$

According to Equations (27)–(29), the steps of gravitational forward modeling of a 3D spherical shell model in the spherical harmonic domain are clear. We first expand the density distributions of each layer into spherical harmonic coefficients, which are then multiplied by the coefficients of corresponding degree and order. Next, the gravitational responses of each layer can be obtained through spherical harmonic synthesis of the result in the first step. Finally, the gravity effects of each layer are summed. For more specificity, Algorithm 1 takes the g_z component as an example to show the detailed procedures of the proposed method.

Algorithm 1: Gravity forward modeling in spherical harmonic domain

Input: 3D density distributions of each layer

1. Do $k = 1$ to N_z , and initialize $g_z = 0$
 2. Calculate ${}^1\bar{\rho}_l^m(r'_k)$ and ${}^2\bar{\rho}_l^m(r'_k)$;
 3. Multiply the factor $\frac{(l+1)}{(2l+1)(l+3)r^{l+2}} \left[\left(r'_k + \frac{1}{2}\Delta r\right)^{l+3} - \left(r'_k - \frac{1}{2}\Delta r\right)^{l+3} \right]$ to step 1;
 4. Spherical harmonic synthesis of the result in step 3 to obtain the $g_z(k)$;
 5. $g_z = g_z + g_z(k)$;
6. End Do

Output: 2D distribution of the g_z component.

Similarly to the procedures of gravitational forward modeling in the Cartesian coordinate system based on FFT [51,52], forward modeling in the spherical harmonic domain also requires spherical harmonic forward and inverse transformations. Therefore, the computational efficiency of spherical harmonic transformations dominates the efficiency of spherical harmonic-based methods. Wiecezorek and Meschede [53] released a software package, SHTOOLS (Version 4.13.1, <https://shtools.github.io/SHTOOLS/> (accessed on 23 September 2023)), in which the Fourier transform software FFTW [54] and the OpenMP parallel program under the “DUCC” library (<https://github.com/litebird/ducc> (accessed on 11 November 2019)) are employed by default to greatly improve the computational efficiency of spherical harmonic expansion and synthesis. In this study, the forward modeling

of a 3D gravitational field in the spherical harmonic domain is implemented under the SHTOOLS software system as expressed in steps 2 and 4 in Algorithm 1. However, we should note that the SHTOOLS is only applicable to the components g_z and g_{zz} , and the computational efficiency for the other components will be much lower.

3. Synthetic Forward Model Tests

3.1. Sphere Shell Model

Here, we first design a simple spherical shell model to test the efficiency of the proposed method compared with the 3D GLQ method. A homogeneous spherical shell with a constant density of 500 kg/m^3 and a thickness of 100 km with the radius ranging from 1638 km to 1738 km , as shown in Figure 2, is discretized into small tesserooids using a set of intervals ranging from 0.25° to 20° and 1 layer in depth. The gravitational fields are calculated and compared on a spherical surface at 10 km above the top surface of the shell model with observation points aligning with the tesserooid mesh. The analytical solution of the gravitational field of this sphere shell model can be calculated using the difference in the gravity responses of two spheres with different radii [27].

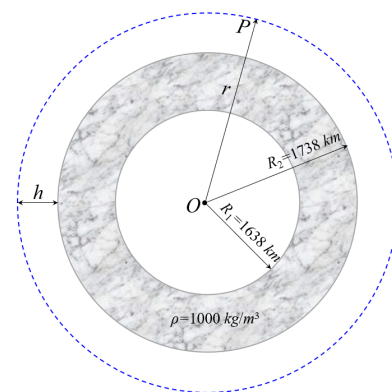


Figure 2. Sketch map of a single-layer global spherical shell model with inner radius $R_1 = 1638 \text{ km}$, outer radius $R_2 = 1738 \text{ km}$, and density $\rho = 1000 \text{ kg/m}^3$. The observation point P is located on the sphere with the height h from the outer sphere.

We calculate the g_z and g_{zz} at different mesh intervals (i.e., different mesh scales) using the methods of Uieda et al. [32], Zhao et al. [5], and the proposed spherical harmonic domain method, with their relative computational times shown in Figure 3. The absolute errors of g_z and g_{zz} in the spatial domain method and our proposed spherical harmonic domain method are shown in Figure 4. In addition, we also show the statistics of the absolute computational time and the maximum relative error of g_z and g_{zz} when the mesh interval is 0.25° (i.e., the discretization is $720 \times 1440 \times 1$) in Table 1.

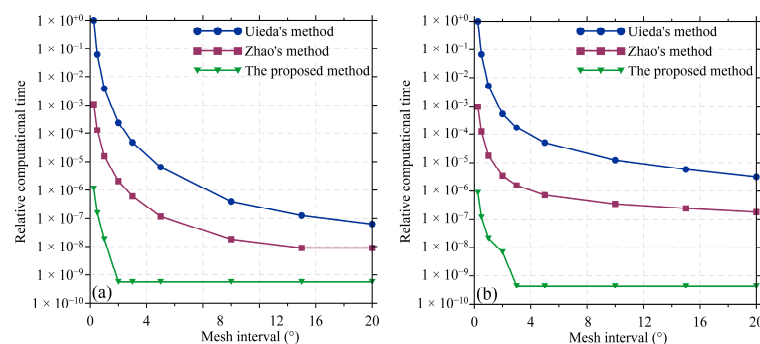


Figure 3. Comparison of relative computational times of (a) g_z and (b) g_{zz} at different mesh intervals in the two spatial domain methods and the proposed method. In the figure, Uieda's method and Zhao's method represent the methods proposed by Uieda et al. [32] and Zhao et al. [5], respectively.

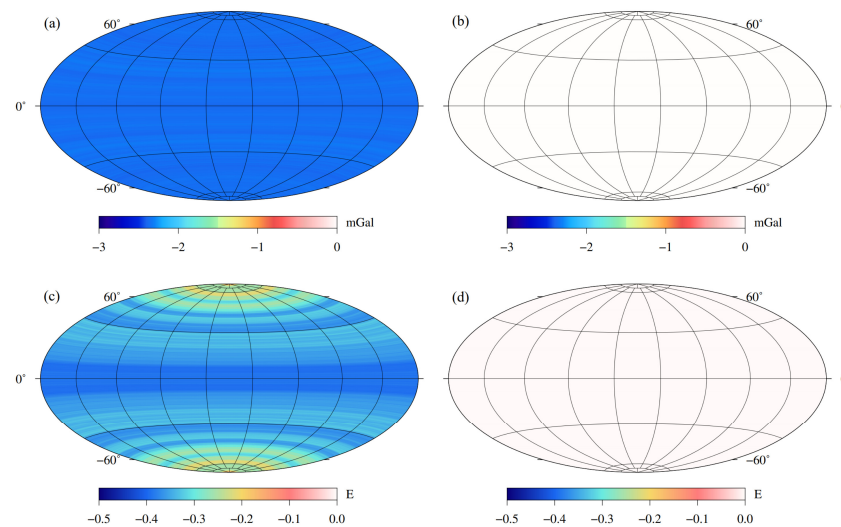


Figure 4. Distribution of absolute error of g_z in (a,b) and g_{zz} in (c,d), where (a,c) are derived from the spatial domain methods and (b,d) are from the proposed spherical harmonic domain method.

Table 1. Statistics of the absolute computational time and the maximum relative errors of g_z and g_{zz} in the two spatial domain methods and the proposed method when the mesh interval is 0.25° . The test was performed on a laptop with an Intel Core i7-11800H CPU and 16 GB RAM.

Methods	Computational Time (s)		Maximum Relative Error (%)	
	g_z	g_{zz}	g_z	g_{zz}
Uieda et al. [32]	1,850,759.53	2,256,446.96	3.12×10^{-2}	5.15×10^{-4}
Zhao et al. [5]	1880.19	2148.41	3.12×10^{-2}	5.15×10^{-4}
The proposed method	1.96	1.96	6.15×10^{-6}	3.38×10^{-6}

It is clearly seen in Figure 3 that the computational time increases rapidly with the decrease in mesh intervals (i.e., with an increase in the total number of tesserooids). Specifically, the computational efficiency achieved by the proposed spherical harmonic domain method is increased by 3 and 6 orders of magnitude compared with the methods of Zhao et al. [5] and Uieda et al. [32], respectively, for the g_z component when the mesh interval is less than 3° (Figure 3a). As for the g_{zz} component, the improvements in computational efficiency are more significant: 4 and 7 orders of magnitude (Figure 3b), respectively. These improvements in computational efficiency are quantified in Table 1, where the absolute computational times obtained by the spatial domain methods are extremely long, while that obtained by the proposed spherical harmonic domain method is acceptable and suitable for global 3D gravity inversion.

As for computational accuracy, Figure 4 shows that the absolute errors of g_z and g_{zz} obtained by both the spatial and spherical harmonic domain methods vary with latitude. It can also be seen that the absolute errors in the proposed spherical harmonic domain method (Figure 4b,d) are lower than those in the spatial domain method (Figure 4a,c). Specifically, the maximum relative error of the g_z component in the proposed spherical harmonic domain method is decreased by four orders of magnitude compared with that in the spatial domain method, as shown in Table 1, while the accuracy for the g_{zz} component improves by about two orders of magnitude.

According to Heck and Seitz [28], the computational accuracy of the spatial domain method based on Taylor series expansion and Gaussian–Legendre integration is usually very low and sometimes even wrong results are obtained when the observation height is close to the surface of a tesseroid. To guarantee a high computational accuracy, it is necessary to finely discretize the tesseroid until the geometric size of the newly meshed tesseroids is smaller than the distance from the geometric center of the meshed tesseroid

to the observation point. Uieda et al. [32] proposed an adaptive discretization algorithm to guarantee that the maximum relative errors are less than 0.1%; however, this is at the cost of a significant increase in computational time. Here, we design a set of experiments to further test the influence of different observation heights on the computational efficiency of the proposed method compared with spatial domain methods. The spherical shell is discretized into 180×360 tesseroids with an equal interval of 1° in both the latitudinal and longitudinal directions. The observation surface has the same discretization as the spherical shell with its elevation ranging from 0.01 to 250 km above the top surface of the spherical shell. The computational times for g_z and g_{zz} at different observation heights are shown in Figure 5 and Table 2.

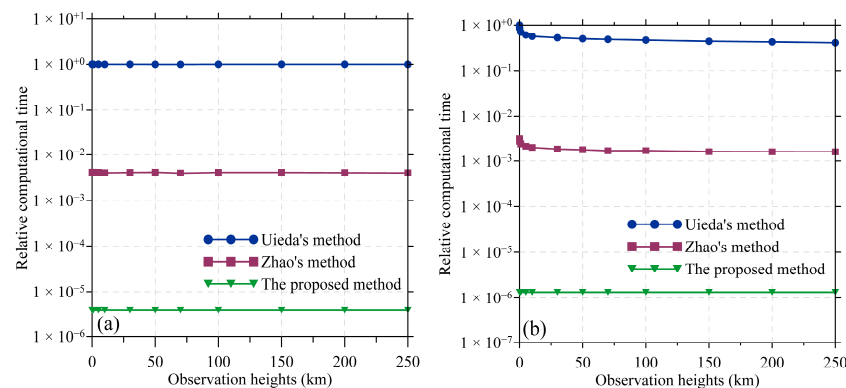


Figure 5. Comparison of relative computational times for (a) g_z and (b) g_{zz} at different observation heights between the two spatial domain methods and the proposed method. In the figure, Uieda's method and Zhao's method represent the methods proposed by Uieda et al. [32] and Zhao et al. [5], respectively.

Table 2. Statistics of the absolute computational time for g_{zz} by the two spatial domain methods and the proposed method at observation heights of 0.01, 0.1, 1.0, 10.0, and 100.0 km.

Observation Heights (km)	Computational Time for g_{zz} (s)		
	Uieda et al. [32]	Zhao et al. [5]	The Proposed Method
0.01	11,631.53	36.81	0.016
0.1	9831.45	32.20	0.016
1.0	8143.89	28.14	0.016
10.0	6588.89	23.33	0.016
100.0	5428.82	19.79	0.016

As can be seen from Figure 5 and Table 2, the computational time for the g_z component of both the spatial domain method and the proposed method hardly changes with the change in observation heights. As for the g_{zz} component, it is obvious that the computational time of the proposed method does not change with the variation in observation heights, while the computational time of the spatial domain methods, especially that of Uieda et al. [32], increases significantly with the decrease in observation heights. In particular, when the observation surface is very close to the tesseroid unit (e.g., the observation height < 0.1 km), the computational time of the g_{zz} component of the spatial domain methods is more than twice as long as that when the observation surface is high (e.g., the observation height > 10 km). For the spatial domain method using the adaptive discretization algorithm to achieve the same computational accuracy (e.g., a maximum relative error less than 0.1%), the number of meshed subdivisions when calculating the g_{zz} component is more than five times that when calculating the g_z component, which leads to a significant increase in the computational time of the g_{zz} component when the observation

surface is low. However, the proposed method's computational time is not affected by the observation height since there is no need for the fine subdivision of tesseroid units.

3.2. Complex Synthetic Model

To further test the effectiveness of the proposed method, we designed a more complex synthetic model. The model is a spherical shell with a thickness of 100 km ranging from 1638 km to 1738 km, which is evenly discretized into $360 \times 720 \times 10$ tesseroids (i.e., with a mesh interval of 0.5° in the horizontal direction and ten layers in depths) with density anomalies varying as follows:

$$\rho(r, \varphi, \lambda) = (1738 - r) \cdot [\sin(2\varphi) \cdot \sin(2\lambda)] \quad (30)$$

The density distribution of this model on a layer with a radius of $r = 1682.5$ km is shown in Figure 6. The gravitational fields are calculated on a spherical surface 10 km above the top surface of this model. Figure 7 shows the g_z and g_{zz} of this model obtained by the spatial domain method, and the differences between it and the proposed spherical harmonic domain method.

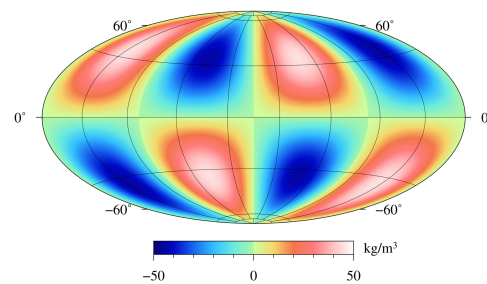


Figure 6. Density distribution of the complex synthetic model at the radius of 1682.5 km.

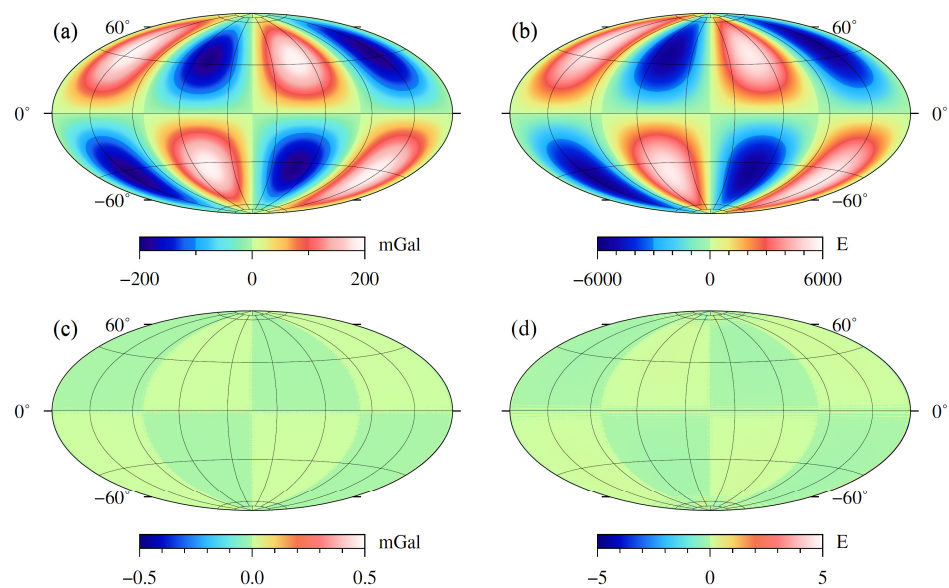


Figure 7. The gravitational responses of g_z (a) and g_{zz} (b) in this complex model derived from the spatial domain methods, with (c,d) showing the differences in g_z and g_{zz} between the two methods, respectively.

Although it is difficult for this complex model to have an analytical solution, the results obtained using the spatial domain method with the adaptive discretization strategy [32] can be regarded as standards for comparison. It can be seen in Figure 7 that the g_z and g_{zz} of the proposed spherical harmonic domain method are very close to (a) and (b) of the spatial domain method, with the greatest differences in g_z (c) and g_{zz} (d) between the

two methods lower than 0.5 mGal and 5.0 E, respectively; this proves the correctness and effectiveness of the proposed method.

4. Application to Lunar Topography Correction

The Bouguer gravity anomaly is a comprehensive reflection of all density anomalies in the subsurface after removing the influence of topography from the free-air gravity anomaly which has been widely used in 3D gravity inversion to reveal the density structure and deformation mechanism of the lithosphere. In addition, the Bouguer gravity anomaly is also the basis for other gravity corrections, such as the residual gravity anomaly that is used for Moho surface inversion by removing the gravity effects of sedimentary layers, crystalline crust layers, and deep mantle from the Bouguer gravity anomaly. Here, we apply the proposed forward method to calculate the gravity response of the lunar topography and then obtain the Bouguer gravity anomaly.

The latest lunar gravitational field model GL1500E [21] provided by the GRAIL mission is used here to calculate the free-air gravity anomaly. The model is expressed in the form of spherical harmonic coefficients, where different degrees and orders represent gravity anomalies with different spatial resolutions [55]. In this study, the free-air gravity anomaly is calculated with truncated degrees of 450 corresponding to a spatial resolution of $0.4^\circ \times 0.4^\circ$ using the spherical harmonic synthesis method [56]. Considering that the observation data are usually extended upwards to a certain height approximately equal to the width of the surface cells in inversions to reduce the influence of noise in data [57], we calculate the free-air gravity anomaly at a height of 10 km above the reference radius of 1738 km, as shown in Figure 8a.

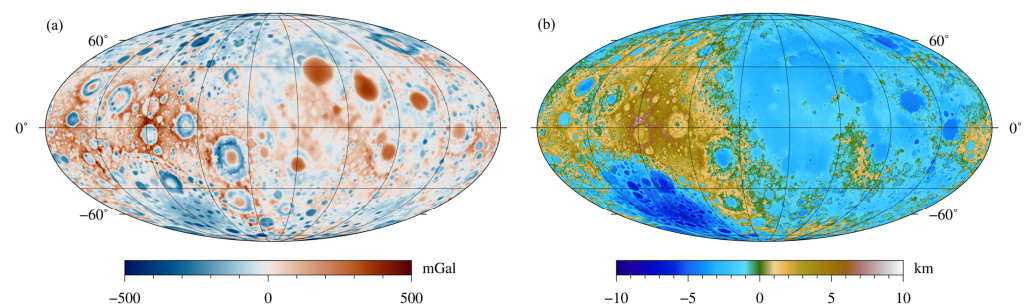


Figure 8. (a) Free-air gravity anomaly at 10 km height derived from the GL1500E model [21] with truncated degrees of 450; (b) topography of the Moon derived from the LRO_LTM05 model [58].

Next, we calculate the gravity effect of the lunar topography. The topography (Figure 8b) of the Moon is derived from the model LRO_LTM05_2050 [58] using the same degrees and orders of spherical harmonic coefficients as the free-air gravity anomaly to maintain consistency. Considering that the proposed forward method is suitable for uniform layered models, which are inconsistent with the undulating terrain model, it is necessary to discretize the terrain into plenty of tesseroids with constant thickness for each layer to reduce the error of fitting the terrain using tesseroids. Figure 8b shows the lunar topography ranges from -9.29 to 9.63 km, so we construct a spherical shell that is slightly higher than and can cover the topography, ranging from -10.00 to 10.00 km, and then this spherical shell is evenly divided into 400 thin layers with an interval of 0.05 km, as shown in Figure 9. Next, it is necessary to judge the relationship between the depth of the geometric center of each tesseroid and the terrain. If the tesseroid is covered by the terrain, the correction density of this tesseroid is assigned as 2560 kg/m^3 (for mountains) or -2560 kg/m^3 (for basins); otherwise, it is 0 kg/m^3 . The topography correction is also calculated on a surface with a radius of 1748 km , consistent with the free-air gravity anomaly. Figure 10 shows the gravity effects of the lunar topography calculated using the spatial domain method, and the differences between it and the proposed method.

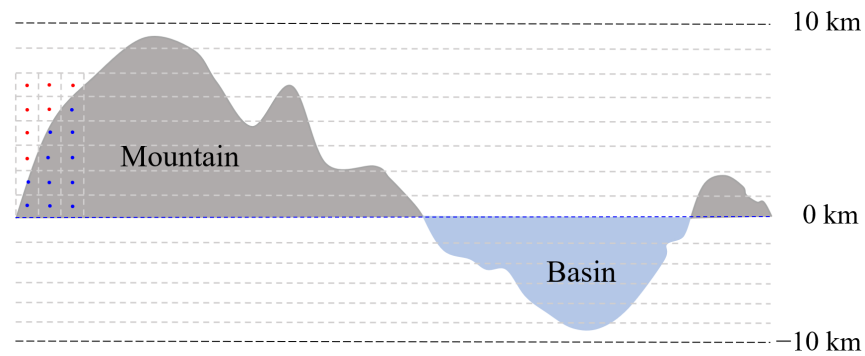


Figure 9. Sketch map of the discretization of the terrain in the radial direction. The dark gray represents mountains with a correction density of 2560 kg/m^3 , and the light blue represents basins with a correction density of -2560 kg/m^3 . Red dots indicate that the tesserooids are outside the terrain with a correction density of 0 kg/m^3 , and blue dots indicate that the tesserooids are inside the terrain.

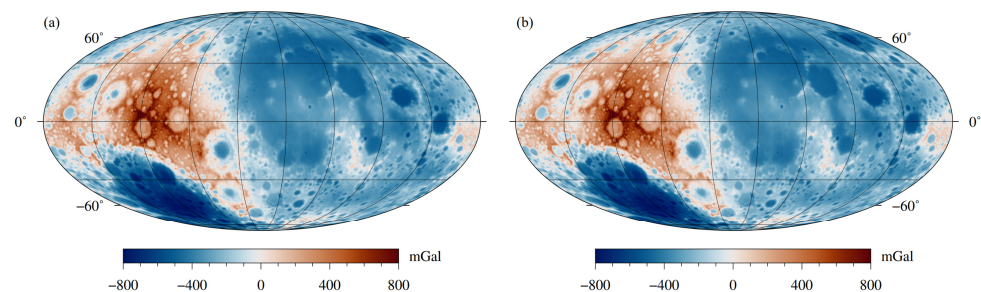


Figure 10. Gravity effects of the lunar topography calculated with (a) the spatial domain method and (b) the proposed spherical harmonic domain method.

Figure 10 shows that the gravity effect of the lunar topography obtained using the proposed method is similar to the result obtained using the spatial domain method, which further proves the effectiveness and practicality of the proposed method. The computational times of the topography corrections in the proposed method and the spatial domain method are 88.69 s and 48,789.98 s, respectively, indicating the high efficiency of the proposed method. However, the advantage of the proposed method in terms of computational efficiency is slightly weakened in this application compared with that in the synthetic example in Section 3. This is because the premise of the proposed method is that all tesserooids of each layer have the same thickness. Therefore, when calculating terrain correction, the entire terrain has to be divided into several thin layers (e.g., 400 layers in this example).

We remove the gravity effect of topography from the free-air gravity anomaly to obtain the Bouguer gravity anomaly. Since the results of topography corrections performed with the spatial domain method and the proposed method are close, we just show the Bouguer gravity anomaly obtained using the proposed method in Figure 11a. The Bouguer gravity anomaly in Figure 11a ranges from -500 mGal to 900 mGal , where the positive high anomalies are mainly distributed in most mascon basins and the South Pole–Aitken basin, while the negative anomalies are distributed on the Highland on the farside of the Moon. It can be found that the Bouguer gravity anomaly in Figure 11a is very close to the Bouguer result of Liang et al. [14], and the only difference between them is that the latter one used spherical harmonic coefficients with degrees and orders less than 60 to calculate the free-air gravity anomaly and the topography correction, which makes its resolution much lower than that of this study. However, the features of the Bouguer gravity anomaly in Figure 11a are very different from the results of Neumann et al. [56] and Zhao et al. [5]. It can be seen that the positive and negative Bouguer anomalies in Figure 11a display obvious asymmetries between the nearside and farside of the Moon as well as between the northern and southern hemispheres. As for the results in Neumann et al. [56] and Zhao et al. [5], as shown in Figure 11b, positive high anomalies are significantly distributed

beneath most mascon basins with less significant negative anomalies scattered around the positive anomalies (i.e., mascon basins). This is because they used 6–450 degrees and orders of spherical harmonic coefficients to calculate the free-air gravity anomaly and the topography correction, where the effect of the hemispheric asymmetry and the South Pole–Aitken basin has been naturally removed by discarding the signal with degrees and orders less than 6.

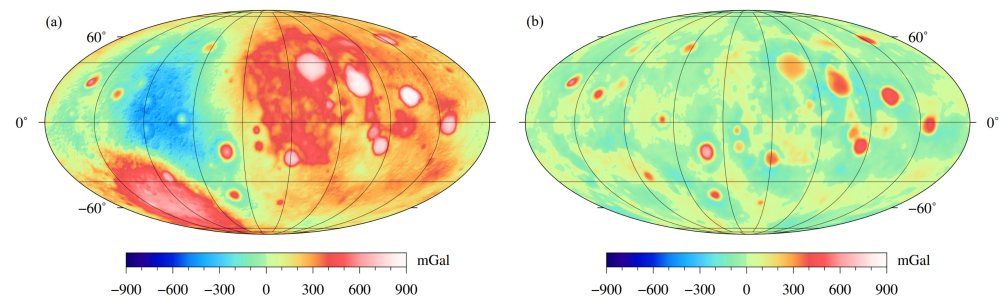


Figure 11. (a) Bouguer gravity anomaly obtained by subtracting the topography correction in Figure 10a from the free-air gravity anomaly in Figure 8a; (b) Bouguer gravity anomaly obtained using 6–450 degrees and orders of spherical harmonic coefficients.

5. Discussion

Topography correction with the spatial domain method essentially uses gravity forward modeling to calculate the gravity response of the terrain observed on a surface at a certain height. In this way, the obtained topography correction contains all degrees and orders of signals. To calculate the Bouguer gravity anomaly with desired degrees and orders, the topography correction should be calculated with the same degrees and orders, which is generally difficult for the spatial domain method. However, this is very easy for spherical harmonic domain methods. As shown in Equations (16)–(25), the gravity effects of the topography within desired degrees and orders are directly calculated through spherical harmonic synthesis.

We note that topography correction is only an application example of the proposed method. More generally, this method can serve as a forward kernel for the iterative inversion of 3D gravitational field in spherical coordinate systems, which can greatly improve the computational efficiency of inversion modeling. As for the commonly used Tikhonov regularization inversion theory, it is necessary to calculate and store the forward kernel matrix, and then calculate the product operation of the kernel matrix and other vectors. However, there is no explicit expression of the kernel matrix in the above equations. Therefore, how to use the proposed forward method for inversion is a problem worthy of consideration. Although the method proposed in this study cannot explicitly calculate the elements of the kernel matrix, it is worth noting that the kernel matrix is only calculated and stored once in the whole iterative inversion process, while the product operation of the kernel matrix and other vectors is performed hundreds of times. Therefore, in the inversion, we still use the equivalent storage strategy to calculate and store the kernel matrix [4], and adopt the proposed high-efficiency forward method to calculate the product operation of the kernel matrix and other vectors (i.e., a complete process of gravity forward modeling).

We should also note that the proposed spherical harmonic domain method is only applicable to global forward modeling since the spherical harmonic functions are defined on the surface of a sphere. However, gravity anomaly and density structures are usually of greater concern for local regions than the global scale. For this case, the spherical cap harmonic analysis will be a good choice for local gravitational fields [59–61]. The spherical harmonic domain method proposed in this study can also be analogized to spherical cap harmonic analysis for local gravity forward and inversion modeling.

Furthermore, the results of gravitational field forward modeling can provide an important reference for the interpretation of remote sensing data and help identify geological structures and abnormal areas in remote sensing images. At the same time, remote sensing

data can also provide necessary surface information for gravitational field forward modeling, such as topographic relief and rock types, so as to improve the accuracy and reliability of forward modeling. In addition, in geological exploration and resource investigation, gravitational field forward modeling and remote sensing technology are often combined to form a comprehensive exploration system. This can greatly improve the efficiency and accuracy of exploration by predicting the location and properties of underground geological bodies through gravitational field forward modeling and combining with remote sensing data to observe and analyze the surface. Therefore, the high-efficiency gravitational forward modeling method proposed in this study provides a foundation for the joint research of gravity and remote sensing.

6. Conclusions

This study presented a high-efficiency gravitational forward modeling method in a spherical coordinate system based on spherical harmonic transform, in which the gravity anomalies and gradient tensors can be expressed as a spherical harmonic synthesis form of spherical harmonic coefficients of 3D density distribution. Two synthetic forward examples demonstrated that the computational efficiency of the proposed method has improved by four orders of magnitude and with a similar level of computational accuracy compared with the optimized 3D GLQ method. The verified gravity forward method is then applied to calculate the topography correction and the Bouguer gravity anomaly of the Moon using the latest high-resolution gravitational field model and topography model. The result shows that the gravity effect of the lunar topography obtained with the proposed method is similar to the result obtained using the spatial domain method. The computational efficiency of the topography correction is relatively lower than the synthetic forward examples because the terrain should be evenly discretized into many thin layers to satisfy the requirement of the proposed forward method. In addition, the proposed method can calculate topography correction and the Bouguer gravity anomaly with any desired degrees and orders, which is generally different for the spatial domain method. However, we should note that the proposed spherical harmonic domain method is only applicable to global gravity forward modeling, which may be analogized and applied to the spherical cap harmonic analysis for local gravity forward and inversion modeling.

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Data Availability Statement: The data and codes for this study are freely available at the website <https://figshare.com/s/ad39b1440639d3e0f631> (accessed on 25 June 2024) [62].

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Conflicts of Interest: The authors declare no conflicts of interest.

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